

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Debugging & Optimization Task

Course Code:	CSA0309	Course Title:	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	Assessment:	Assessment Tool 2 – Debugging & Optimisation Task
Weightage:	35% of CIA	Total Marks:	100
CO2 Statement:	<i>Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.</i>		
Student Name:	N.Adi Narayana	Reg. No.:	192525346
Section / Batch:	AIML	Date:	22/07/2026

Code of Conduct

I _____ N .Adi Narayana _____
(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the Student: __N .Adi_____

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Search Data Structure, Stack	Binary Search, Stack Operations	1 - 5	25

Linked List Data Structure	List Operations	6 – 10	25
Queue Data Structures	Queue Operations	11 - 15	25
Queue Data Structures, Stack Notations	Queue Operations, Expression Evaluation	16 - 20	25
GRAND TOTAL:		100 Marks	

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Debugging Accuracy	20	All errors corrected; code runs flawlessly	Most errors corrected; minor issues remain	Partial correction; code runs with issues	Code fails or major errors persist
Error Identification	30	Accurately identifies all logical, syntax, and edge-case errors	Identifies most major errors	Identifies some errors but misses key issues	Struggles to identify errors
Code Quality & Readability	20	Clean, modular, well documented, follows best practices	Mostly clean and readable	Some structure but inconsistent	Poorly written, hard to understand
Explanation & Justification	20	Clear, logical, and well justified reasoning with complexity analysis	Good explanation with some justification	Limited explanation; lacks depth	No or incorrect explanation
Testing & Validation	10	Comprehensive test cases including edge cases	Adequate test cases	Minimal testing	No testing provided
Faculty In-charge		Course Coordinator		Program Director	
Signature: _____		Signature: _____		Signature: _____	
Date: _____		Date: _____		Date: _____	

DEBUGGING & OPTIMIZATION TASK (Set 1)

1)

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Q6
Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List Data Structures
5 marks

Q9
Linked List
5 marks

Q10
Queue
5 marks

10/10 answered

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 int binarySearch(int arr[100], int n, int key) {
4     int low = 0, high = n - 1;
5     while (low <= high) {
6         int mid = (low + high) / 2;
7         if (arr[mid] == key)
8             return mid;
9         else if (arr[mid] < key)
10            low = mid + 1;
11        else
12            high = mid - 1;
13    }
14    return -1;
15 }
```

Test Input

```
7
2 4 6 8 10 12 14
8
```

Output

Compilation Error:
C:/CCPPComp/CCPP4/bin/./lib/gcc/
x86_64-w64-
mingw32/10.3.0/./.././../x86_64-
w64-mingw32/bin/ld.exe:
C:/CCPPComp/comp4/bin/./lib/gcc/

Pending manual review

Submitted

2)

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Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack
5 marks

Q6
Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List Data Structures
5 marks

5 marks

Find the error in the stack deletion operation using the below code.

```
int pop0 {
    if(top == 0) {
        printf("Underflow");
        return -1;
    }
    return stack[top--];
}
```

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 int stack[100];
3 int top = -1;
4 int pop() {
5     if(top == -1) {
6         printf("Underflow");
7         return -1;
8     }
9     return stack[top--];
10 }
11 int main() {
12     return 0;
13 }
```

Test Input

stdin input (optional)

Output

3)

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack
5 marks

Q6
Linked List
5 marks

Q7
Stack
5 marks

5 marks

Q3: Stack

Check the snippet and identify the logical error.

```
int stack[5];
int top;
void init0 {
    top = 0;
}
```

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 int stack[5];
2 int top;
3
4 void init() {
5     top = -1;
6 }
```

Test Input

stdin input (optional)

Output

4)

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

QUESTIONS

Q1 Binary Search 5 marks

Q2 Stack 5 marks

Q3 Stack 5 marks

Q4 Stack 5 marks

Q5 Stack 5 marks

Q6 Linked List 5 marks

Q7

Q4 Stack 5 marks

Identify the errors in the following snippet.

```

if(ch == ')') {
    while(stack[top] != ')') {
        postfix[j++] = pop();
    }
    pop();
}

```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```

1 if (ch == ')') {
2     while (top >= 0 && stack[top] != ')') {
3         postfix[j++] = pop();
4     }
5     if (top >= 0 && stack[top] == ')') {
6         pop(); // remove ')' from stack
7     }
8 }

```

Test Input
(a*(b+c))

Output

5)

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QUESTIONS

Q2 Stack 5 marks

Q3 Stack 5 marks

Q4 Stack 5 marks

Q5 Stack 5 marks

Q6 Linked List 5 marks

Q7 Stack 5 marks

Q8 Linked List Data Structures 5 marks

Q2 Stack 5 marks

Find the error in the stack deletion operation, using the following code snippet.

```

int pop() {
    if(top == 0) {
        printf("Underflow");
        return -1;
    }
    return stack[top--];
}

```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```

1 #include <stdio.h>
2 int stack[100];
3 int top = -1;
4 int pop() {
5     if(top == -1) {
6         printf("Underflow");
7         return -1;
8     }
9     return stack[top--];
10 }
11 int main() {
12     return 0;
13 }

```

Test Input
stdin input (optional)

Output

6)

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

QUESTIONS

Q1 Binary Search 5 marks

Q2 Stack 5 marks

Q3 Stack 5 marks

Q4 Stack 5 marks

Q5 Stack 5 marks

Q6 Linked List 5 marks

Q7

Q6 Linked List 5 marks

What is the issue?
temp = head
while temp.next != NULL:
print(temp.data)
temp = temp.next

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```

1 temp = head
2 while temp is not None:
3     print(temp.data)
4     temp = temp.next

```

Test Input
stdin input (optional)

Output

7)

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QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack 5 marks
- Q6 Linked List 5 marks
- Q7 Stack 5 marks

Q7 Stack 5 marks

Point out the error.
 POP(stack):
 if top == 0:
 print("Underflow")
 else:
 return stack[top--]

★ **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 POP(stack):
2     if top == -1:
3         print("Underflow")
4     else:
5         return stack[top--]
```

Test Input
 stdin input (optional)

Output

8)

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1) Back

QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack 5 marks
- Q6 Linked List 5 marks
- Q7 Stack 5 marks

Q8 Linked List Data Structures 5 marks

Identify the errors in the following code
 while current != NULL:
 current.next = prev
 prev = current
 current = current.next

★ **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 prev = None
2 current = head
3
4 while current is not None:
5     next_node = current.next
6     current.next = prev
7     prev = current
8     current = next_node
```

Test Input
 stdin input (optional)

Output

9)

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1) Back

QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack 5 marks
- Q6 Linked List 5 marks
- Q7 Stack 5 marks

Q9 Linked List 5 marks

What is the error in the below code?
 new.next = head
 head = new

★ **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 new.next = head
2 head = new
```

Test Input
 stdin input (optional)

Output

10)



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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)
Back

QUESTIONS

- Q1** Binary Search 5 marks
- Q2** Stack 5 marks
- Q3** Stack 5 marks
- Q4** Stack 5 marks
- Q5** Stack 5 marks
- Q6** Linked List 5 marks
- Q7** Stack 5 marks

Q10 Queue 5 marks

What is missing in the below pseudocode?

```

DEQUEUE(Q):
if front == -1:
print("Underflow")
else:
return Q[front++]

```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```

1 DEQUEUE(Q):
2 if front == -1:
3     print("Underflow")
4 else:
5     x = Q[front]
6     front = front + 1
7     if front > rear:
8         front = -1
9         rear = -1

```

Test Input

Output

DEBUGGING & OPTIMIZATION TASK (Set 2)

1)



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- Q4** Queue 5 marks
- Q5** Queue 5 marks
- Q6** Doubly Linked List 5 marks
- Q7** Stack 5 marks
- Q8** Stack 5 marks
- Q9** Circular Queue 5 marks
- Q10** Stack 5 marks

```

push(result,
)
}

```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```

1 int i = 0;
2
3 while (expression[i] != '\0') {
4     if (isdigit(expression[i])) {
5         push(expression[i] - '0');
6     } else {
7         int op1 = pop();
8         int op2 = pop();
9         int result;
10
11         if (expression[i] == '+') result = op2 + op1;
12         else if (expression[i] == '-') result = op2 - op1;
13         else if (expression[i] == '*') result = op2 * op1;
14         else if (expression[i] == '/') result = op2 / op1;
15
16         push(result);
17     }
18     i++;
19 }

```

Test Input

Output

2)

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Q1 Stack Notations - Postfix Expression 5 marks

Q2 Circular Queue 5 marks

Q3 Binary Search 5 marks

Q4 Queue 5 marks

Q5 Queue 5 marks

Q6 Doubly Linked List 5 marks

Q7 Stack 5 marks

Q2 Circular Queue 5 marks

Identify the bug in this Circular Queue dequeue logic.

```
int dequeue() {
    if (front == -1) return -1;
    int data = queue[front];
    front = (front + 1) % MAX;
    return data;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 int dequeue() {
2     if (front == -1) return -1;
3
4     int data = queue[front];
5
6     if (front == rear) {
7         front = rear - 1;
8     } else {
9         front = (front + 1) % MAX;
10    }
11    return data;
12 }
```

Test Input

stdin input (optional)

Output

3)

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Q1 Stack Notations - Postfix Expression 5 marks

Q2 Circular Queue 5 marks

Q3 Binary Search 5 marks

Q4 Queue 5 marks

Q5 Queue 5 marks

Q6 Doubly Linked List 5 marks

Q7 Stack 5 marks

Q3 Binary Search 5 marks

Identify the bug in this Recursive Binary Search.

```
int binarySearch(int arr[], int l, int r, int x) {
    int mid = l + (r - l) / 2;
    if (arr[mid] == x) return mid;
    if (arr[mid] > x) return binarySearch(arr, l, mid, x);
    return binarySearch(arr, mid, r, x);
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 int binarySearch(int arr[], int l, int r, int x) {
2     int mid = l + (r - l) / 2;
3
4     if (l > r) return -1;
5     if (arr[mid] == x) return mid;
6
7     if (arr[mid] > x)
8         return binarySearch(arr, l, mid - 1, x);
9     else
10        return binarySearch(arr, mid + 1, r, x);
11 }
```

Test Input

stdin input (optional)

Output

4)

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Q1 Stack Notations - Postfix Expression 5 marks

Q2 Circular Queue 5 marks

Q3 Binary Search 5 marks

Q4 Queue 5 marks

Q5 Queue 5 marks

Q6 Doubly Linked List 5 marks

Q7 Stack 5 marks

Q4 Queue 5 marks

Find the error in this Deque insertFront operation.

```
void insertFront(int data) {
    if (front == 0) front = MAX - 1;
    else front = front - 1;
    deque[front] = data;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 void insertFront(int data) {
2     if ((front == 0 && rear == MAX - 1) || (front == rear + 1)) {
3         printf("Deque is Full\n");
4         return;
5     }
6     if (front == -1) {
7         front = 0;
8         rear = 0;
9     } else if (front == 0) {
10        front = MAX - 1;
11    } else {
12        front = front - 1;
13    }
```

Test Input

stdin input (optional)

Output

5)



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QUESTIONS

Q1
Stack Notations - Postfix Expression
5 marks

Q2
Circular Queue
5 marks

Q3
Binary Search
5 marks

Q4
Queue
5 marks

Q5
Queue
5 marks

Q6
Doubly Linked List
5 marks

Q7
Stack
5 marks

Q5 Queue 5 marks

Identify the error in the Queue enqueue logic.

```
void enqueue(int data) {
    if (rear == MAX) return;
    queue[rear++] = data;
}
```

★ **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #define MAX 100
2
3 int queue[MAX];
4 int front = -1, rear = -1;
5
6 void enqueue(int data) {
7     if (rear >= MAX - 1) {
8         printf("Queue is full\n");
9         return;
10    }
11    if (front == -1) {
12        front = 0;
13    }
14}
```

Test Input

Output

6)



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QUESTIONS

Q1
Stack Notations - Postfix Expression
5 marks

Q2
Circular Queue
5 marks

Q3
Binary Search
5 marks

Q4
Queue
5 marks

Q5
Queue
5 marks

Q6
Doubly Linked List
5 marks

Q7
Stack
5 marks

Q6 Doubly Linked List 5 marks

Identify the error in this Doubly Linked List node deletion.

```
void deleteNode(Node* del) {
    if (del->prev != NULL)
        del->prev->next = del->next;
    if (del->next != NULL)
        del->next->prev = del->prev;
}
```

★ **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 void deleteNode(Node* del) {
2     if (del == NULL)
3         return;
4
5     if (del->prev != NULL)
6         del->prev->next = del->next;
7
8     if (del->next != NULL)
9         del->next->prev = del->prev;
10
11    free(del); // Free memory to avoid memory leak
12}
```

Test Input

Output

7)



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QUESTIONS

Q1
Stack Notations - Postfix Expression
5 marks

Q2
Circular Queue
5 marks

Q3
Binary Search
5 marks

Q4
Queue
5 marks

Q5
Queue
5 marks

Q6
Doubly Linked List
5 marks

Q7
Stack
5 marks

Q7 Stack 5 marks

Identify the issue in the below pseudocode.

POP():
if top == 0 → Underflow

★ **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 POP() {
2     if top == -1:
3         print "Underflow"
4         return
5     else:
6         value = stack[top]
7         top = top - 1
8         return value
9 }
```

Test Input

Output

8)

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

Q1
Stack Notations - Postfix Expression
5 marks

Q2
Circular Queue
5 marks

Q3
Binary Search
5 marks

Q4
Queue
5 marks

Q5
Queue
5 marks

Q6
Doubly Linked List
5 marks

Q8 Stack 5 marks

Point out the error.
if top == MAX:
Overflow

✪ **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```

1 MAX = 5
2 top = -1
3 stack = [0]*MAX
4
5 def push(x):
6     global top
7     if top == MAX - 1:
8         print("Overflow")
9     else:
10        top += 1
11        stack[top] = x
12
13 # test
14 push(5)
          
```

Test Input

stdin input (optional)

Output

9)

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

Q1
Stack Notations - Postfix Expression
5 marks

Q2
Circular Queue
5 marks

Q3
Binary Search
5 marks

Q4
Queue
5 marks

Q5
Queue
5 marks

Q6
Doubly Linked List
5 marks

Q9 Circular Queue 5 marks

Identify the issue in the following snippet for Circular Queue in Traffic Signal.

```

rear = (rear + 1) % MAX;
queue[rear] = vehicle;
if(rear == front){
    printf("Queue Full");
}
          
```

✪ **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```

1 int next = (rear + 1) % MAX;
2
3 if (next == front) {
4     printf("Queue Full");
5 } else {
6     rear = next;
7     queue[rear] = vehicle;
8 }
          
```

Test Input

stdin input (optional)

Output

10)



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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

Q1
Stack Notations - Postfix
Expression
5 marks

Q2
Circular Queue
5 marks

Q3
Binary Search
5 marks

Q4
Queue
5 marks

Q5
Queue
5 marks

Q6
Doubly Linked List
5 marks

Q10 Stack5 marks

Point out the error?
PEEK(stack):
return stack[top+1]

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 PEEK(stack):  
2     return stack[top]
```

Test Input

stdin input (optional)

Output